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MILITARY SPECIFICATION

PREAMPLIFIER, TRAVELING WAVE TUBE

AM-6965/DLQ-3B

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, test, manufacture and acceptance requirements for the AM-6965/DLQ-3B traveling wave tube (TWT) preamplifier, herein referred to as the preamplifier.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 Preservations, Methods of.

Beneficial comments (recommendations, additions, deletions, and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5865

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MIL-E-5400	Electronic Equipment, Aerospace, General Specification for.
MIL-T-5422	Testing, Environment, Airborne Electronic and Associated Equipment.
MIL-I-6181	Interference Control Requirements, Aircraft Equipment.
MIL-E-17555	Electronic and Electrical Equipment Accessories and Repair Parts; Packaging and Packing of.
MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for.
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment, Including Ground Support Equipment.
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities.
MIL-I-81219	Indicator, Elapsed Time, Electrochemical.
MIL-C-81703	Connector, Electric, Circular, Miniature, Rack and Panel or Push-Pull Coupling, Environment Resisting.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-280	Definitions of Item Levels, Item Interchangeability, Models and Related Terms.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-704	Aircraft Electrical Power Characteristics.
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.

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MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-882	System Safety Program Requirements.
MIL-STD-965	Parts Control Program.
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-45662	Calibration System Requirements.
MS-3134	Connector, Receptacle, Electric, Solder Type, Jam Nut Mounting, Push-Pull Coupling, Series 1 (Class E, P, J. and H).
MS 3137	Connector, Plug, Electric, Solder Type, Push-Pull Connector.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

1356AS216	Preamplifier, Traveling Wave Tube AM-6965/DLQ-3B.

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item description. The preamplifier covered by this specification is a TWT used to amplify and serrodyne modulate radio frequency (RF) signals existing in the microwave region of electromagnetic energy.

3.1.1 Components. The preamplifier unit shall, within the meaning of MIL-STD-280, contain the following separate assemblies.

a. TWT assembly.
b. Power supply assembly.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

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3.3 Interface.

3.3.1 External electrical and RF. The external electrical interfaces for the preamplifier shall be provided through connectors J6001, J6002, J6003, and J6004, and located as shown on Drawing 1356AS216. Connector type and function shall be as specified in Table I.

TABLE I. Electrical and RF interface types and functions.

Reference Designation	Connector Type	Function
J6001	MS3134E19-4P	Input Power
J6002	Type SMA	RF Output
J6003	Type SMA	RF Input
J6004	Type SMA	Helix Modulation

3.3.2 External mechanical. The external mechanical interface dimensions shall be as shown on Drawing 1356AS216. The mechanical interface shall enable direct installation into government selected electronic platforms using existing mounting holes.

3.4 Materials, processes, and parts. In the selection of materials, processes, and parts, fulfillment of major design objectives for the intended application specified herein shall be the prime consideration. In doing so, the following shall govern:

a. Microwave technology shall be considered and microwave items shall conform to requirements specified herein.

b. Materials, processes, and parts requirements shall be as specified in MIL-E-5400.

c. Contractor-designed (peculiar) microwave items shall meet the general intent as specified in MIL-E-5400.

d. Approval for the use of nonstandard materials, processes, and parts shall be obtained as specified in MIL-E-5400.

3.5 Characteristics.

3.5.1 Performance requirements. The preamplifier shall, when employed in conjunction with other units of the AN/DLQ-3B Electronic Countermeasures Set, meet the performance requirements specified herein at the conditions of 4.3. Electron tubes shall be interchangeable between preamplifiers with no supply voltage changes.

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3.5.1.1 Frequency range. The frequency range of the preamplifier shall be from 4 to 12 gigahertz (GHz).

3.5.1.2 RF output power. The RF output power of the preamplifier shall be 21 decibels above one milliwatt (dBm) for the 4 to 4.5 and 11.5 to 12 GHz ranges and 23 dBm for the 4.5 to 11.5 GHz range.

3.5.1.3 Termination mismatch. The preamplifier shall meet specified frequency range and RF output power with an input and output voltage standing wave ratio (VSWR) of 2.0:1 nominal. The preamplifier shall not be damaged when operating into load with VSWRs up to and including 2.5:1 over the frequency range and at any phase angle.

3.5.1.4 Input power.

3.5.1.4.1 Input voltage. The preamplifier shall operate as required when supplied a power source as specified in Table II and Figures 9 and 10 of MIL-STD-704 except the Steady State Voltage shall be 28.0 ±3.0 Volts direct current (Vdc).

3.5.1.4.2 Current drain. The preamplifier current drain shall be not greater than 1.5 amperes (A).

3.5.1.5 Warm-up time. Within 90 seconds after application of primary power, the set shall achieve specified output power and all other performance requirements.

3.5.1.6 Modulation.

3.5.1.6.1 Serrodyne modulation. The preamplifier shall produce serrodyne modulated outputs throughout its 4 to 12 GHz operating range. Sidebands shall be suppressed not less than 20 decibels (dB) below primary signal levels throughout the operating frequency range. Application of sawtooth voltages between 20 and 80 volts peak-to-peak having frequencies between 50 Hz and 1 MHz shall produce serrodyne modulation with a sideband suppression of not less than 20 dB at all frequencies in the 4 to 12 GHz range.

3.5.1.6.2 Intermodulation. With normal dc operating voltage applied and two equal RF input signals applied at a level which causes an RF output of 18 dBm, the third-order intermodulation products shall be down not less than 15 dB in the 4 to 12 GHz operating range.

3.5.1.7 Signal gain. The small signal gain shall be greater than +40 dB for the 4 to 4.5 and 11.5 to 12 GHz ranges and +43 dB for the 4.5 to 11.5 GHz range. Maximum gain shall be not greater than 63 dB throughout the 4 to 12 GHz operating range.

3.5.1.8 Duty cycle. The preamplifier shall be capable of 100 percent duty cycle.

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3.5.1.9 Spectral purity. Unless otherwise specified in the contract or purchase order (see 6.2.1), all residual noise components within ±100 kHz of the carrier shall be down not less than 45 dB from the carrier.

3.5.1.10 Broadband tube noise. The broadband tube noise power output shall be not greater than +7 dBm.

3.5.1.11 Thermal protection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the preamplifier shall be equipped with automatic, resettable thermal overload protection.

3.5.1.12 Harmonic and spurious emissions. In band and out of band harmonic and spurious emissions shall be supressed not less than 30 dB from the carrier level.

3.5.2 Physical characteristics.

3.5.2.1 Conformance to documents. The preamplifier shall be as specified in Drawing 1356AS216 and shall meet all the requirements of this specification and the documents referenced to the extent specified herein.

3.5.2.2 Dimensions. The length, width, and height of the preamplifier shall be as specified in Drawing 1356AS216.

3.5.2.3 External adjustments. No external adjustments shall be available on the preamplifier.

3.5.2.4 Weight. The weight of the preamplifier shall be not greater than 11 pounds.

3.5.2.5 Transportation and storage. Preamplifiers shall be capable of being transported by normally available common commercial carriers (air and land) and military carriers (land, sea and air) without damage or adverse impact on operating characteristics. Shipping and storage containers/packaging shall be adequate for storage in facilities normally available to U.S. military units.

3.6 Environmental conditions. The preamplifier shall meet all requirements specified herein prior to, during, and after the following environmental conditions, separately or any combination thereof. The preamplifier shall not exhibit any temporary or permanent degradation as the result of exposure to any of the following environments:

a. Temperature-altitude: Barometric pressure reduced to the equivalent of 50,000 feet altitude and a temperature of -40°C.

b. Vibration: Random vibration at 10 to 2000 Hz and ±10 g RMS.

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c. Shock: Acceleration pulses at 25 g levels with a duration of 11 ms in all axes in any direction.

d. Humidity: An atmosphere containing 85 percent or greater relative humdity at temperature ranging from +28°C to +71°C.

e. Explosion: An explosive atmosphere containing a computed air/fuel ratio of 13:1 at 10,000 feet altitude increments from sea level to 50,000 feet.

f. Storage temperature: Temperatures ranging from -54°C to +71°C.

g. Operating temperature: Temperatures ranging from -40°C to +71°C.

h. Thermal shock: Temperatures ranging from -54°C to +71°C.

3.7 Shelf life. The preamplifier shall be designed to function and operate as required herein following a storage period of 5 years in normally available government storage facilities at temperatures varying from -54°C to +71°C and relative humidities varying from 5 percent to 95 percent.

3.8 Reliability. The preamplifier shall have a mean-time-between-failure (MTBF) of not less than 100 hours when tested as specified in MIL-STD-781, Test Level F (modified to limit testing between -40°C and +71°C) and using the acceptance criteria specified in Test Plan V-C.

3.9 Design and construction.

3.9.1 Materials. All materials used in the construction of the preamplifier shall be as specified in MIL-STD-454, Requirement 3, Flammable Materials; Requirement 4, Fungus-Inert Materials; Requirement 15, Ferrous Alloys, Corrosion Resistance and Requirement 16, Dissimilar Metals.

3.9.2 Electromagnetic interference. Unless otherwise specified in the contract or purchase order (see 6.2.1), the preamplifier shall not generate EMI signals either through radiation, power line conduction, amplitude or phase modulation of the output signal nor through spikes fed back to inputs selected or not selected. Shielding of the preamplifier from stray or operating environment EMI shall be provided as specified in MIL-I-6181 and MIL-STD-461.

3.9.3 Nameplate and marking. The nameplate shall be as specified in Drawing 1356AS216. Interchangeable parts and assemblies shall be legibly marked. All markings shall be as specified in MIL-STD-130.

3.9.3.1 Nomenclature and serial numbers. The contractor shall prepare requests for nomenclature and serial number assignments as specified in MIL-N-18307 (see 6.2.2).

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3.9.4 Interchangeability. The preamplifier unit, with separately attached or integrated power supply assembly, shall be interchangeable as specified in MIL-STD-454, Requirement 7. Electrical and electronic component parts shall conform to the requirements specified in MIL-STD-965.

3.9.5 Safety. The preamplifier produced as specified in this specification shall present no safety hazard to operating personnel. Provisions of MIL-STD-882 shall be complied with.

3.9.5.1 Safety interlock. If external cabling connects the power supply assembly to the TWT, the power supply shall have an interlocking circuit whereby primary DC power is applied after the high voltage connection is made to the TWT and removed prior to disconnection.

3.9.6 Human performance/human engineering. The preamplifier shall be constructed and configured to minimize human error during installation, operation, removal and maintenance. MIL-STD-1472 and MIL-H-46855 shall be used by the contractor to meet human engineering and performance requirements.

3.9.7 Standards of manufacture.

3.9.7.1 Primary DC power connector. The primary DC electrical power connector for the preamplifier shall be as specified in MIL-C-81703, Class E and shall mate with MS3137E19-4S connector.

3.9.7.2 RF coaxial connector. The preamplifier RF coaxial connectors shall be front bulkhead mounted, female receptacle, type SMA.

3.9.7.3 TWT power connector. If external cabling is employed, TWT power shall be connected utilizing J6005, an AMP Inc. 862825-1 connector and mating receptacle or an equal and interchangeable connector.

3.9.7.4 Annunciators. The preamplifier shall incorporate illuminated annunciators to indicate the operational conditions specified in Table II.

TABLE II. Annunciator operational conditions.

Annunciator	Color	Condition When Illuminated
a. PWR ON	Yellow	+28 Vdc input power is applied.
b. STBY 1/	Amber	Power applied, TWT not timed out.
c. OPR 1/	Green	Power applied, TWT timed out.
d. FAULT 1/	Red	No TWT beam current.

1/ Mutually exclusive - no simultaneous illumination.

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3.9.7.5 Elapsed time indicators. The power supply and TWT assemblies, if individually packaged, shall incorporate elapsed time indicators as specified in MIL-I-81219 and Drawing 1356AS216.

3.9.7.6 Power time delay. The preamplifier shall incorporate a time delay feature for application of high voltages to the TWT.

3.10 Workmanship. The preamplifier, including all parts and accessories, shall be constructed and finished in a manner that will ensure compliance with all requirements of this document. Unless otherwise specified herein, fabrication and assembly practices shall be as specified in MIL-STD-454, Requirement 9.

3.10.1 Soldering practices. Soldering practices shall be as specified in MIL-STD-454, Requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall assure product conformance to the requirements, inspections, and tests specified herein. The contractor's quality assurance program shall be planned and used in a manner to achieve levels of quality and reliability commensurate with the intended application specified herein. The contractor shall notify the procuring activity of planned testing not less than 10 days prior to testing.

4.1.1.1 Process control procedures. Process control procedures shall be prepared when necessary to supplement applicable process specifications to provide detailed performance and control methods. These procedures shall document the preparation; fabrication details; conditions to be maintained during each phase of the process; the methods of verifying the adequacy of processing materials, solutions, equipment and their associated control parameters, including statistical quality control plans where applicable and the required records to indicate the results of such inspection and process verification. The contractor's quality organization shall review the written procedures for process controls and conduct audits to determine that the actual operations conform with approved methods and procedures.

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4.1.2 Test plans. Unless otherwise specified in the contract or purchase order (see 6.2.1), detailed test plans as specified in MIL-T-18303 shall be prepared by the contractor prior to the commencement of testing (see 6.2.2).

4.1.3 Test data. Unless otherwise specified in the contract or purchase order (see 6.2.1), the detailed test data from both the first article tests and acceptance tests shall be prepared by the contractor (see 6.2.2). Test data for each preamplifier offered for acceptance shall be as specified in MIL-STD-831. This data shall identify all rejections from acceptance tests and corrective actions taken.

4.1.4 Failure records. Unless otherwise specified in the contract or purchase order (see 6.2.1), failure analysis and proposed corrective action shall be prepared by the contractor (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.5).
b. Quality conformance inspections (see 4.6).

(1) Acceptance tests (see 4.6.1).
(2) Reliability tests (see 4.6.2).
(3) Manufacturing burn-in test (see 4.6.3).

4.3 Inspection conditions. Unless otherwise specified in the applicable test procedure, all tests and inspections shall be conducted under the following conditions:

a. Temperature (ambient): $+23^{\circ} \pm 10^{\circ}C$

b. Atmospheric Pressure: 725 +50/-115 mm of mercury (28.5 +2.0/-4.5 inches of mercury)

c. Humidity: 10 to 90 percent

4.3.1 External power. The external power shall be $+28.0 \pm 3.0$ Vdc applied to J6001 and shall simulate airborne vehicle power transfer, when specified in the applicable test procedure.

4.3.2 Test equipment accuracy. Unless otherwise specified herein, all test equipment used shall have an accuracy within 20 percent of the tolerance for the variable to be measured. In some cases where an open-ended minimum or maximum requirement is given, the test equipment used shall be accurate to within 5 percent of that requirement.

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4.3.2.1 Maintenance and calibration. All test equipment shall be maintained and calibrated as specified in MIL-STD-45662. Certification to this effect prior to all inspections and tests shall be prepared by the contractor (see 6.2.2).

4.4 First article sample. Unless otherwise specified in the contract or purchase order (see 6.2.1), a first article sample consisting of two preamplifiers shall be delivered to a designated testing activity specified in the contract or purchase order (see 6.2.1), for first article inspection. The first article sample shall be manufactured using the same methods, materials, processes and procedures proposed for production.

4.4.1 First article approval. Approval of the first article sample shall be by the procuring activity upon satisfactory completion of all tests. The procuring activity reserves the right to require an additional sample and test if any preamplifier fails to meet any requirement specified herein (see 6.2.1). Any production by the contractor before acceptance of the first article sample will be at the contractor's risk. Subsequent to first article approval, the contractor shall not change materials, processes, or procedures without prior approval of the procuring activity (see 6.2.2).

4.5 First article tests. The first article sample shall be subjected to the inspections and tests of Table III and shall be performed in the sequence specified therein. The contractor shall prepare and include test reports (see 6.2.2).

4.5.1 First article failure analysis and corrective action. Unless otherwise specified in the contract or purchase order (see 6.2.1), the procuring activity shall be notified within one working day of a first article test failure and its probable cause. Procuring activity approval is required before implementation of any corrective action in first article products (see 6.2.2).

4.5.2 Disposition of first article test sample. Unless otherwise specified in the contract or purchase order (see 6.2.1), preamplifiers which have been subjected to the first article tests shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the preamplifiers for acceptance.

4.6 Quality conformance tests.

4.6.1 Acceptance tests. Acceptance tests shall be conducted on each preamplifier offered for acceptance and shall consist of the tests specified in Table III, except for the warm-up test of 4.7.2.1. The first four of each five preamplifiers produced shall be tested for conformance with items a through g of 4.7.2.1. The fifth shall be tested for conformance with items a through j of 4.7.2.1.

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TABLE III. First article and quality conformance tests.

Examination or Test	Requirement Paragraph	Method Paragraph	
		First Article	Acceptance
1. Visual examinations			
a. Dimensions	3.5.2.2	4.7.1.1	4.7.1.1
b. Weight	3.5.2.4	4.7.1.1	4.7.1.1
c. Nameplate and markings	3.9.3	4.7.1.3	4.7.1.3
d. Interchangeability	3.9.4	4.7.1.4	4.7.1.4
e. Workmanship	3.10	4.7.1.5	4.7.1.5
2. Manufacturing burn-in test		4.6.3	
3. Performance			
a. Warm-up	3.5.1.5	4.7.2.1	4.7.2.1
b. Primary power voltage variation (25.0 Vdc)	3.5.1.4.1	4.7.2.4	
c. Primary power voltage variation (31.0 Vdc)	3.5.1.4.1	4.7.2.5	
d. Termination mismatch	3.5.1.3	4.7.2.2	
e. Time meter	3.9.7.5	4.7.2.3	
f. Electromagnetic interference	3.9.2	4.7.2.6	
4. Environmental	3.6	4.7.2.13	
5. Reliability	3.8	4.6.2	

4.6.1.1 Acceptance test data. The contractor shall prepare acceptance test data sheets (see 6.2.2) for each preamplifier offered for acceptance.

4.6.2 Reliability tests. Test plan V-C as specified in MIL-STD-781 shall be used to determine acceptance or rejection of the reliability test. Theta (θ) shall be 100 hours. All time accumulated in 4.6.1 shall be counted in determining whether an accept or reject condition exists. The contract or purchase order shall specify whether, upon equipment failure prior to reaching an accept or reject decision, the selected first article unit or units shall be repaired to the satisfaction of the Government and subjected to further reliability testing or whether a subsequent first article unit or units shall be randomly selected (see 6.2.1). The contractor shall prepare a reliability test plan, test reports, and a failure summary and analysis report (see 6.2.2).

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4.6.2.1 Reliability failure criteria. In addition to the failure criteria as specified in MIL-STD-781, a preamplifier shall be considered to have failed when any of the monitored characteristics perform outside the specified limits for that characteristic. Preventive maintenance measures or adjustments shall not be performed upon the preamplifiers during the period of the test.

4.6.2.2 Reliability data and test reports. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall prepare reliability test data and reports, including failure records, for each preamplifier subjected to reliability testing (see 6.2.2). The data and reports shall be as specified in MIL-STD-781 and shall be prepared for review not later than 10 working days after completion of reliability testing.

4.6.3 Manufacturing burn-in test. Each preamplifier shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the preamplifier. This test shall be deleted if the reliability test includes a test on each preamplifier which requires at least ten hours of preamplifier on time. The test shall be run at ambient temperature and humidity conditions. Vibration shall be at any non-resonant frequency between 20 and 30 Hz at a level of ± 3 gs. The preamplifier shall be vibrated (without vibration oscillators) for a period of ten minutes prior to the beginning of the ten hour period of operation. Where feasible, the preamplifier shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the preamplifier in two directions, the vertical direction shall be used. During the ten hour period of operation following the ten minute vibration period, the preamplifier shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future preamplifiers. A record shall be kept of all failures. The ten hour period specified above may be composed of two five hour periods to conform with standard working hours.

4.7 Inspection and test methods. Details of methods and procedures shall be as specified in the approved test plan (see 4.1.2).

4.7.1 Visual examinations.

4.7.1.1 Physical characteristics. The preamplifier shall be inspected for conformance with 3.5.2.

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4.7.1.2 Materials, processes, and parts. The preamplifier shall be inspected for conformance with 3.4.

4.7.1.3 Nameplate and marking. The preamplifier shall be inspected for conformance with 3.9.3.

4.7.1.4 Interchangeability. The preamplifier shall be examined for conformance with 3.9.4.

4.7.1.5 Workmanship. The preamplifier shall be examined for conformance with 3.10 and 3.10.1.

4.7.2 Performance tests. Unless otherwise specified in the contract (see 6.2.1), performance tests shall be performed to ensure that the preamplifier shall operate as specified in 3.5.1. Input power shall be as specified in 3.5.1.4 and the performance shall be continuously monitored during the conduct of tests.

4.7.2.1 Warm-up test. The preamplifier shall be operated under the provisions and conditions specified in 4.3. The input power applied shall be as specified in 3.5.1.4 and the warm-up time shall be as specified in 3.5.1.5. The preamplifier shall be operated for a period of 30 minutes and the following shall be observed:

a. Frequency range (see 4.7.2.8).
b. RF output power (see 4.7.2.8).
c. Serrodyne modulation (see 4.7.2.9).
d. Signal gain (see 4.7.2.8).
e. Broadband tube noise (see 4.7.2.11).
f. Harmonic and spurious emissions (see 4.7.2.12).
g. Spectral purity (see 4.7.2.7).
h. Intermodulation (see 4.7.2.10).

4.7.2.2 Termination mismatch. The preamplifier shall be connected to a mismatched load having a VSWR of 2:1 with phase angle between the resistive and the reactive load components varied between zero and 360 degrees. Input power shall be 28.0 $\pm$3.0 Vdc and the RF TUNE shall be adjusted to 6.0 GHz and 9.5 GHz and the following observed to determine if the preamplifier meets the requirements specified in 3.5.1.3:

a. Frequency range (see 4.7.2.8).
b. RF output power (see 4.7.2.8).

4.7.2.3 Time meter test. The preamplifier's time meter shall be tested under simulated operational conditions with running time observed and meet the requirements specified in 3.9.7.5.

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4.7.2.4 Primary power voltage variation (+25.0 +0.1 -0.0 Vdc) test. The primary power voltage shall be adjusted to +25.0 +0.1 -0.0 Vdc and the following observed:

a. Frequency range (see 4.7.2.8).
b. RF output power (see 4.7.2.8).

4.7.2.5 Primary power voltage variation (+31.0 +0.0 -0.1 Vdc) test. The primary power shall be adjusted to +31.0 +0.0 -0.1 Vdc and the following observed:

a. Frequency range (see 4.7.2.8).
b. RF output power (see 4.7.2.8).

4.7.2.6 Electromagnetic interference test. The power lead conducted emission test CE01 and the radiated emission electric field test RE02 as specified in MIL-STD-462 shall be performed to ensure that the preamplifier meets the requirements specified in 3.9.2. When specified in the contract or purchase order (see 6.2.1), EMI test results shall be prepared by the contractor for each production lot (see 6.2.2.).

4.7.2.7 Spectral purity test. The spectral purity of the preamplifier shall be tested to verify conformance as specified in 3.5.1.9 in 2 GHz steps from 4.0 GHz to 12.0 GHz.

4.7.2.8 Frequency range, RF output power, and signal gain. The frequency range, RF output power, and signal gain of the preamplifier shall be tested to verify conformance specified in 3.5.1.1, 3.5.1.2, and 3.5.1.7 respectively, by testing RF output power and signal gain from 4.0 to 12.0 GHz in 1 GHz steps.

4.7.2.9 Serrodyne modulation. The serrodyne modulation of the preamplifier shall be tested to verify conformance specified in 3.5.1.6.1 from 4.0 to 12.0 GHz in 2 GHz steps.

4.7.2.10 Intermodulation. The intermodulation of the preamplifier shall be tested to verify conformance specified in 3.5.1.6.2 from 4.0 to 12.0 GHz in 2 GHz steps.

4.7.2.11 Broadband tube noise. The broadband tube noise of the preamplifier shall be tested to verify conformance specified in 3.5.1.10.

4.7.2.12 Harmonic and spurious emissions. The harmonic and spurious emissions of the preamplifier shall be tested to verify conformance specified in 3.5.1.12 from 4.0 to 12.0 GHz in 2 GHz steps.

4.7.2.13 Environmental tests. The preamplifier shall conform to requirements for Class 1 equipments operating in environments specified in 3.6 when subjected to testing as specified in Table II of MIL-T-5422, except that salt-fog testing is not required.

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4.7.2.13.1 Operational environmental tests. The preamplifier's performance shall be tested and meet the requirements specified in 3.5.1 during and after exposure to the following environments to verify conformance specified in 3.6.

4.7.2.13.1.1 Temperature and altitude test. The preamplifier shall be tested as specified in the temperature-altitude test as specified in MIL-T-5422 Table II, as modified herein by Table IV.

TABLE IV. Modified MIL-T-5422 temperature-altitude tests.

Step	Temperature (degree C)	Altitude (feet)	Time (hours)	Input Voltage	Operating/ Nonoperating
1	-40	ATM 1/	2	--	N.O.
2	-35	ATM	--	25.0	OP
3	-40	50,000	--	31.0	OP
4	-10	ATM	--	31.0	OP
5	+71	ATM	--	--	N.O.
6	+71	ATM	2	31.0	OP
7	OMIT				
8	OMIT				
9	+44	50,000	1.25	31.0	OP
9A	AMB 2/	AMB	2	--	N.O
9C	+44	50,000	1.25	31.0	OP
10	+58	35,000	2	31.0	OP
11	OMIT				
12	OMIT				
13	OMIT				

1/ ATM equals atmosphere
2/ AMB equals ambient

4.7.2.13.1.2 Vibration test. The preamplifier shall be tested in accordance with the vibration test as specified in MIL-T-5422, Curve I and Curve II as specified in Figure 5.

4.7.2.13.1.3 Shock test. The preamplifier shall be tested in accordance with the shock test as specified in MIL-T-5422, except as modified herein to no greater than 25 g.

4.7.2.13.1.4 Explosion test. The preamplifier shall be tested in accordance with the explosion test specified in MIL-T-5422.

4.7.2.13.2 Nonoperational environmental tests. The preamplifier performance shall be tested and meet the requirements specified in 3.5.1 after exposure to the following environments.

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4.7.2.13.2.1 Thermal shock test. The preamplifier shall be tested in accordance with the thermal shock test as specified in MIL-T-5422.

4.7.2.13.2.2 Humidity test. The preamplifier shall be tested in accordance with the humidity test as specified in MIL-T-5422, except that two continuous cycles shall be used instead of 10.

4.8 Preshipment inspection. Prior to shipment, inspection shall ensure that preservation, packaging, packing, and marking for shipment are in accordance with the requirements specified in Section 5.

5. PACKAGING

5.1 Preservation-packaging, and packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), the level of preservation-packaging, and packing shall be as specified in MIL-E-17555 and MIL-STD-794, Level C. The method of preservation for level C shall be determined in accordance with the selection chart in Appendix D specified in MIL-STD-794 and the method of packaging shall be as specified in MIL-P-116.

5.2 Marking.

5.2.1 Special marking. Special marking shall be as specified in the contract or purchase order (see 6.2.1).

5.2.2 Normal marking. Unit packages and shipping containers shall be marked as specified in the requirements specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The preamplifier is an operating component of an electronic warfare (EW) countermeasures unit for installation in aerial targets and piloted aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Spectral purity, if different than specified (see 3.5.1.9).

c. Thermal protection, if different than specified (see 3.5.1.11).

d. Electromagnetic interference, if different than specified (see 3.9.2).

e. Responsibility for inspection (see 4.1).

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f. Test plans, if different than specified (see 4.1.2).

g. Test data, if different than specified (see 4.1.3).

h. Requirement for failure records (see 4.1.4).

i. First article sample, if different than specified (see 4.4).

j. Designated testing activity for first article inspection (see 4.4).

k. Additional samples and tests, if required (see 4.4.1).

l. Notification time of first article failure, if different than specified (see 4.5.1).

m. Disposition of first article sample, if different than specified (see 4.5.2).

n. Repair or selection of new first article sample for reliability testing (see 4.6.3 and Table III).

o. Reliability data and test reports, if different than specified (see 4.6.2.2).

p. Performance tests, if different than specified (see 4.7.2).

q. EMI test results, if different than specified (see 4.7.2.6).

r. Level of preservation, packaging, and packing, if different than specified (see 5.1).

s. Special markings, if required (see 5.2.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirement title	Applicable DID No.	Option
3.9.3.1	Requests, Nomenclature, and Identification Assignment and Approval	DI-E-21336B	

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Paragraph No.	Data requirement title	Applicable DID No.	Option
4.1.2	Test Procedures	UDI-T-21347	
4.1.3, 4.6.1.1	Records, inspections & tests	UDI-T-22706A	
4.1.4, 4.5.1	Failure Analysis and Corrective Action Report	DI-R-5299C	
4.3.2.1	Equipment Calibration Procedures	DI-R-7065	
4.4.1	Engineering Change Proposals (ECPs), Requests for Waivers and Deviations (Long Form)	DI-E-2037	
4.1.3, 4.5	Test Report, First Article	UDI-T-23790	
4.6.2	Plan Reliability Test	DI-R-7033	
4.6.2, 4.6.2.2	Reports, Reliability Test and Demonstration	DI-R-7034	
4.6.2	Failure Summary and Analysis Report	DI-R-7041	
4.7.2.6	Electromagnetic Interference Test Report	DI-R-7062	

(Data item descriptions related to this specification, and identified in section 6, will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of two preamplifiers. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for examinations, test and approval of the documents first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity

Navy - AS

(Project 5865-N040)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-85433A(AS)	Preamplifier, Traveling Wave Tube AM-6965/DLQ-3B

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

Naval Air Engineering Center
Engineering Specifications and Standards Dept.
(CSD), Code 93
Lakehurst, NJ 08733

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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